

MICHIGAN DEPARTMENT OF NATURAL RESOURCES & ENVIRONMENT

INTEROFFICE COMMUNICATION

TO: Tiffany J. Myers, Permits Section, Water Resources Division

FROM: Doyle Brunsen, Permits Section, Water Resources Division

DATE: January 2, 2013

SUBJECT: Cadillac WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257
Water Quality-Based Effluent Limits (WQBEL)

We have completed our evaluation for water quality-based effluent limits (WQBELs) for the reissuance of the NPDES permit for the Cadillac WWTP and the new use permit for the Haring Township WWTP. The discharges from these facilities are within two miles of each other and have been evaluated together.

The Cadillac WWTP currently continuously discharges 3.2 million gallons per day (MGD) (4.93 cubic feet per second (cfs)) of treated municipal wastewater to the Clam River (NE $\frac{1}{4}$, NE $\frac{1}{4}$, Section 33, R9W, T22N) in Wexford County. The Haring Township WWTP is a new continuous discharge of 0.135 MGD (0.21 (cfs)) of treated municipal wastewater to the Clam River (NE $\frac{1}{4}$, NW $\frac{1}{4}$, Section 27, R9W, T22N) in Wexford County.

The Clam River is protected for warmwater fish from Lake Cadillac to County Line Road and coldwater fish downstream of County Line Road as well as other indigenous aquatic life and wildlife, fish consumption, partial body contact recreation, total body contact recreation from May through October, agriculture, navigation, industrial water supply, and public water supply at the point of intake.

The monthly 50% and 95% exceedance flows, in cfs, for the Clam River at this location are as follows (HSDSU #8068):

	<u>JAN.</u>	<u>FEB.</u>	<u>MAR.</u>	<u>APR.</u>	<u>MAY</u>	<u>JUNE</u>
95%	4.6	4.3	8.1	12	6.3	3.7
50%	16	15	34	73	31	11
	<u>JULY</u>	<u>AUG.</u>	<u>SEP.</u>	<u>OCT.</u>	<u>NOV.</u>	<u>DEC.</u>
95%	2.7	2.8	2.9	3.4	5.2	5.3
50%	6.5	5.5	5.7	10	18	18

These flows have changed from the previous flows used for recommendations.

RECOMMENDATIONS

The recommendations found in tables 1 and 2 have changed since the last reissuance of the Cadillac WWTP NPDES permit. The new limits reflect the changes in background flow and the addition of the Haring Township discharge. The discharge monitoring reports from the Cadillac WWTP indicate the past effluent quality is sufficient to meet the more restrictive limits.

The recommended limits for the Cadillac WWTP are found in table 1. Our model predicts that the coldwater D.O. standard in the Clam River will be adequately protected if the Cadillac WWTP

(December through March) and Haring Township WWTP (all year) effluent is of a quality that meets the federal secondary treatment definition. An effluent D.O. limit of 3.0 mg/l minimum is recommended for the entire year based on the acute D.O. toxicity criterion for warmwater fish. Effluent quality has been modeled as daily maximum of 60 mg/l CBOD₅ and 30 mg/l ammonia-nitrogen, and as a daily minimum of 3.0 mg/l for D.O. These recommendations meet the coldwater dissolved oxygen (D.O.) water quality standard (WQS) of 7 mg/l as a daily minimum at County Line Road. Ammonia toxicity WQBELs are based on the 0.029 mg/l un-ionized ammonia toxicity criterion for coldwater streams.

We evaluated the effects of the discharge on the Clam River using a Multi-reach Streeter-Phelps D.O. model. We modeled the effect of the discharge on the receiving stream based on a study (time of passage, reaeration, flow measurements) done in 1983 (Synk, MDEQ Report #025120). The diurnal variation (daily average minus daily minimum concentrations) of 0.75 mg/l and the background D.O. are based on site specific data collected in 2006 (Brunsen, MI/DEQ/WB-08/Draft).

The recommendations contained herein are based on WQS. Treatment methods or economic concerns have not been addressed. We do not imply in these recommendations that other methods or considerations should not be used in deciding the permit limits.

cc: Brian Jankowski, Cadillac District Supervisor, Water Resources Division
Effluent Limit File, Water Resources Division

TABLE 1
January 2, 2013

FACILITY: Cadillac WWTP
DISCHARGE: Continuous
FACILITY DESIGN FLOW: 3.2 MGD (4.93 cfs)
RECEIVING STREAM: Clam River

EFFLUENT					
PARAMETER	TIME PERIOD	CONC mg/l	MASS lb/d	COMMENT	RATIONALE
D.O.	MAY 1-SEP 30	6.0		DAILY MINIMUM	D.O. STANDARD
	OCT 1-NOV 30	6.0		DAILY MINIMUM	D.O. STANDARD
	DEC 1-MAR 31	3.0		DAILY MINIMUM	ACUTE D.O. TOXICITY
	APR 1-APR 30	5.0		DAILY MINIMUM	D.O. STANDARD
CBOD ₅	MAY 1-SEP 30	28	750	DAILY MAXIMUM	D.O. STANDARD
	OCT 1-NOV 30	39	1,000	DAILY MAXIMUM	D.O. STANDARD
	APR 1-APR 30	32	850	DAILY MAXIMUM	D.O. STANDARD
NH ₃ -N	MAY 1-SEP 30	1.0	27	30-DAY AVERAGE	CHRONIC TOXICITY

OCT 1- NOV 30	2.2	59	30-DAY AVERAGE	CHRONIC TOXICITY
DEC 1- MAR 31	3.7	99	30-DAY AVERAGE	CHRONIC TOXICITY
APR 1- APR 30	4.8	130	30-DAY AVERAGE	CHRONIC TOXICITY

TABLE 2
January 2, 2013

FACILITY: Haring Township WWTP
DISCHARGE: Continuous
FACILITY DESIGN FLOW: 0.135 MGD (0.21 cfs)
RECEIVING STREAM: Clam River

PARAMETER	TIME PERIOD	EFFLUENT		COMMENT	RATIONALE
		CONC mg/l	MASS lb/d		
D.O.	ALL YEAR	3.0		DAILY MINIMUM	ACUTE D.O. TOXICITY
NH ₃ -N	MAY 1- SEP 30	2.0	2.2	30-DAY AVERAGE	CHRONIC TOXICITY
	OCT 1- NOV 30	3.0	3.4	30-DAY AVERAGE	CHRONIC TOXICITY
	DEC 1- MAR 31	7.0	7.9	30-DAY AVERAGE	CHRONIC TOXICITY
	APR 1- APR 30	4.5	5.1	30-DAY AVERAGE	CHRONIC TOXICITY